

Isaac Pressman

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EDUCATION

California Polytechnic State University, San Luis Obispo (Cal Poly)

San Luis Obispo, CA

Bachelor of Science in Computer Science – GPA 3.93

Sep. 2024 – May 2028

Relevant Coursework: Data Structures, Computer Organization, Project-Based Object Oriented Programming, Systems Programming, Discrete Structures, Design and Analysis of Algorithms, Intro to Computer Security, Theory of Computation, Intro to Software Engineering, Database Systems

EXPERIENCE

Software Engineer Intern

June 2026 – Aug. 2026

Coherent

Fremont, CA

- Designed I2C-based CDB capability validation checks for CMIS transceiver firmware upgrades, preventing unsupported upgrade commands and reducing the risk of bricked modules or incompatible image deployments.

Software Engineer Intern

Jan. 2026 – June 2026

Teradyne

Agoura Hills, CA

- Architected an LLM-driven test generation framework in C# for **embedded firmware** validation, increasing coverage from **85% to 99%** and uncovering 100+ driver and control-path edge cases.
- Engineered a **gRPC bridge** between .NET 4.6 and .NET 10 to enable Playwright-driven automation, powering the full nightly smoke test suite for a React-based scan pattern visualization tool.
- Refactored **15,000+ lines** of legacy C# test infrastructure and system components to eliminate duplicated logic across hardware control paths, reducing code redundancy by **73%** and improving long-term maintainability.
- Contributed to IG-XL scan test functionality on the Teradyne UltraFlex platform, supporting **ATPG stuck-at fault** detection via serial scan pattern shift and response capture.

Research Software Developer

Aug. 2024 – Sep. 2025

UC Berkeley Optometry

Berkeley, CA

- Developed a Python (OpenCV, NumPy) computer vision pipeline to measure tear film lipid thickness with **13nm** accuracy, matching **\$20k+** commercial physical systems.
- Built automated segmentation system for 100+ clinical videos with pupil/iris detection across **500+ frames** each.
- Generated a physics-based RGB lookup table using Fresnel equations and CIE color models, enabling pixel-level lipid layer thickness maps and per-frame statistical analysis.
- Reduced video analysis time from **5 minutes to 10 seconds** per video, enabling 10× faster research throughput while maintaining research-grade accuracy.

PROJECTS

Music Listening and Release Tracker | *FastAPI (Python), React Native, PostgreSQL, Spotify Web API*

- Built FastAPI backend with **dual authentication (Spotify OAuth + JWT)** and automatic token refresh, enabling seamless session continuity without re-authentication prompts.
- Engineered fault-tolerant data synchronization service with retry logic and exponential backoff, achieving **99% successful sync rate** despite intermittent Spotify API failures.

Dungeon Crawler | *Unity, C#*

- Built **A*-based pathfinding** and FSM-driven enemy AI with dynamic obstacle avoidance and behavior trees.
- Implemented a save/load system persisting **full game state** across sessions using Unity's JsonUtility.
- Shipped a complete dungeon crawler on itch.io (500+ downloads), owning the **core gameplay systems** end-to-end across a 5-person cross-functional team.

TECHNICAL SKILLS

Languages: Java, Python (OpenCV, NumPy, SciPy, Matplotlib), C, C#, JavaScript, TypeScript, SQL (PostgreSQL), HTML/CSS/SCSS

Frameworks: Django, .NET, React, ReactNative, Node.js, Express.js, FastAPI, Unity

Technologies: Git, Docker, AWS (S3, EC2, IAM), Playwright, gRPC, I2C